Spatial mapping of greenhouse gases using laser absorption spectrometers at local scales of interest

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ABSTRACT

Over the past two years a new system capable of measuring the 2-D spatial distribution of atmospheric CO_2 over areas on the order of 1 km^2 and time scales of a few minutes, has been developed and demonstrated. The Greenhouse gas Laser Imaging Tomography Experiment (GreenLITE) - developed under a cooperative agreement with the National Energy Technology Laboratory (NETL) of the U.S. Department of Energy (DOE) - attempts to improve monitoring capabilities of Ground Carbon Storage (GCS) sites. GreenLITE sensors are based on an intensity modulated continuous wave (IM-CW) approach developed at ITT (now part of Harris Corp.) in 2004. The GreenLITE system recently completed a remote deployment of nearly 4,000 hours at a GCS site in Illinois. It provided continuous, real-time spatial distribution maps of CO_2 via an open web-based interface from February to August 2015. In early 2015 we began work on a new implementation of GreenLITE capable of providing similar measurements over a 25 km^2 area and are planning to test the system over a 5 km range late summer 2015. If successful the system will be deployed in an urban environment late 2015, demonstrating the utility of real-time 2-D spatial mapping of CO_2 concentrations at this scale. This paper will review the concept for this new measurement capability, including results from the 1 km system. Ultimately, the measurement concept can be adapted to other greenhouse gases such as CH_4 and NO_2.

Keywords: Carbon, Laser Absorption Spectroscopy, Tomography

1 INTRODUCTION

The current GreenLITE system is designed to provide the quantitative measurement of atmospheric CO_2 concentrations in a horizontal plane about 1.5 m above the surface, while also providing estimates of spatially resolved concentrations over large areas (1 km^2). This is a unique capability for GCS applications. For comparison, the current state-of-the-art for GCS atmospheric monitoring consists of lidar, *in situ* optical sensors, natural and introduced tracers, and eddy covariance (EC) [1].

In situ optical sensors rely on non-dispersive absorption of IR radiation, usually in a cavity that is temperature controlled, and often after drying the sample to avoid impacts from atmospheric water vapor. These sensors are commercially available and provide precise (~0.1 ppm) measurements of concentration at a single point. They can also provide very accurate data when coupled with appropriate calibration systems using known quantity gas mixtures. Other less expensive sensors are available in exchange for accuracy and precision. Chemical sensors are also available, typically suffering from thermal drifts and contamination from other chemical interactions; they are useful for inexpensive threshold warning systems where only very high concentrations need to be identified for human safety. All *in situ* style sensors only provide a measurement at a given point in space, making it difficult to cover large areas of interest for GCS integrity monitoring.

Natural and introduced tracers can enhance the ability to detect a GCS leak by marking sequestered CO_2 with a tracer. Tracers may make it easier to identify the source of a GCS leak versus naturally varying backgrounds of CO_2 directly. However, they often require specialized equipment for evaluating flask samples that are not commonly located at the GCS site. Another challenge for tracer detection is that the tracer compound may behave differently than the CO_2 due to interaction with other substances or different diffusion rates in the atmosphere, causing the CO_2 release to be misinterpreted.

Eddy covariance uses a combination of sensors to measure rapidly changing vertical and horizontal winds along with rapidly changing CO_2, temperature, humidity, or other trace gases of interest. It relies on a fluid dynamics approach where the atmosphere is assumed to be entirely made up of eddies of varying sizes within a horizontal wind flow [2]. By measuring the up and down parts of the eddy over time along with the local gas concentration and horizontal flow one can determine the net flux at the location of the sensor. It relies on two key assumptions: 1) density fluctuations are assumed negligible, and 2) mean vertical flow is assumed negligible for horizontal homogeneous terrain. The validity of these assumptions depends on the site location and terrain. There are several other assumptions that are made and the analysis of EC data is very complex [3]. It has been shown as a valid approach for determining surface to air fluxes over fairly large areas, but the assumptions need to be satisfied and it is not clear if it can be applied accurately for a wide range of locations and atmospheric conditions. The footprint of an EC measurement is extrapolated using models and the direct flux measurements at the location of the EC tower combined with the horizontal wind flow [4]. Unlike EC, the GreenLITE system, described in Section 2, directly measures the CO_2 density over the entire footprint along multiple chords. The data from the overlapping chords is used in a sparsely sampled tomography approach to get an estimate of the 2-D concentration across the area of interest, as discussed in Section 2.2.2. Differencing the 2-D reconstructions as a function of time can give an estimate of the change in CO_2 concentration with time over the area, which is an estimate of the flux across that boundary. However, the current implementation does not include vertical wind speeds and cannot distinguish between the directions of the flux.

Recently, DOE/NETL supported the development of a compact, eye-safe scanning differential absorption lidar (DIAL) [5] instrument for measuring airborne CO_2 in a horizontal plane. The lidar approach, such as that demonstrated by Repasky from Montana State University (MSU) [6] provides the closest corollary to the GreenLITE measurement, but is distinctly different than the IM-CW laser absorption spectroscopy (LAS) approach implemented in GreenLITE. Both approaches rely on the differential absorption approach, exploiting the difference in atmospheric transmission due to absorption of two closely-spaced, narrow laser lines while extinction due to other processes (primarily scattering) is common to both wavelengths. However, the MSU lidar uses atmospheric backscatter to get the spatially resolved information and, as a result, has a million times lower reflectance than the reflectors used for GreenLITE. This difference causes the DIAL system to be more complex and require larger optics and specialty detectors. Although the MSU instrument showed general agreement with an independent LI-COR system in recent tests, it was only operational at night and showed no correlation to the temporal variability seen by the LI-COR used for comparisons.

Recent testing of the GreenLITE system at the Zero Emissions Research and Technology site in Bozeman, MT [7], included a similar test using a LI-COR based system in the field of regard of the GreenLITE system and comparing it to the average CO_2 measured by GreenLITE. The two instruments showed a correlation coefficient over several hours of >0.933 over multiple tests on multiple days and as high as 0.98 for a 23 hour continuous period during those tests. This represents a significant performance improvement in the sensitivity of this approach and is what is required for an open path laser system trying to detect potential GCS leaks when operated at practical heights above the surface. Prior laser-based open path measurements were also demonstrated by the MSU team [8] which picked up the signature of a controlled ground release but were located only 20 cm above the surface and could not practically be used to monitor a large area where weeds, traffic, and other operations require the measurement to be further from the surface.

Although the original GreenLITE system was designed for GCS monitoring and reporting, similar needs exist for other applications such as urban area monitoring. Currently a number of towers with *in situ* instruments, and other point measurements, are being used to help constrain models, along with inventories, by placing them both upwind and downwind of an urban area [9] [10] [11]. Placing towers within the city proves difficult due to the large local fluctuations that can be seen when measuring at a single point. Scaling the GreenLITE system to areas of 25 or 30 km^2 can enable detailed measurements within the city that are not as sensitive to small scale local variations, and that can provide spatial information for the CO_2 concentrations to better constrain the models. Additionally scaling of the system supports a wider range of GCS applications such as enhanced oil recovery (EOR) fields. Similar challenges exist for measurements of CH_4 at many different source locations from large feedlots to hydraulic fracturing ("fracking") fields. The GreenLITE system conversion to CH_4 is straightforward and is as easy as changing the laser sources and updating the algorithms for the CH_4 spectroscopy.

2 GREENLITE SYSTEM

The GreenLITE system includes both data collection and data processing components. The data collection portion includes transceivers, retroreflectors, and software necessary to operate the transceivers and facilitate data collection. The data processing portion includes algorithms to convert measured differential transmission to integrated CO_2 column concentration and estimate 2-D tomographic reconstructions of CO_2 concentration fields, as well as a web-based user interface for real time monitoring. The following two sections provide an overview of these two aspects.

2.1 GreenLITE Data Collection

The GreenLITE sensor is based on the IM-CW approach [12], which allows multiple wavelengths of light to be transmitted and received simultaneously. The narrow band lasers are set to transmit light on at least two wavelengths, which correspond to an absorption feature of the molecule to be measured. One laser is placed at a wavelength with high absorption and another laser is placed in a region of little absorption. The difference in absorption between the two wavelengths enables the number density of the particular molecule to be determined, assuming the laser and absorption cross-sections are known. The original concept was developed by Dobbs, et al, in 2001 [13] [14] as a method for measuring CO_2 sources and sinks from space. In 2004, ITT Space Systems, now part of Harris Corporation, developed the Multifunctional Fiber Laser Lidar (MFLL) as an airborne demonstrator. NASA Langley Research Center has been evaluating the airborne system since 2005 and has demonstrated that MFLL provides both high-accuracy and high-precision measurements as compared to independent measurements traceable to World Meteorological Organization (WMO) standards. The MFLL instrument was recently selected by NASA Headquarters as a primary instrument for Atmospheric Carbon and Transport – America (ACT-America), a NASA Earth Venture Suborbital mission led by Dr. Ken Davis of Pennsylvania State University.

In 2011 and 2012, Harris Corporation developed a ground-based version of the MFLL instrument to demonstrate a novel concept for making the measurement in transmission and enabling a geostationary solution for obtaining high-accuracy and high temporal resolution CO_2 measurements not currently offered by any planned or existing space-based remote sensing systems [15]. The GreenLITE system leveraged the best aspects of the prior work, resulting in a robust and capable design.

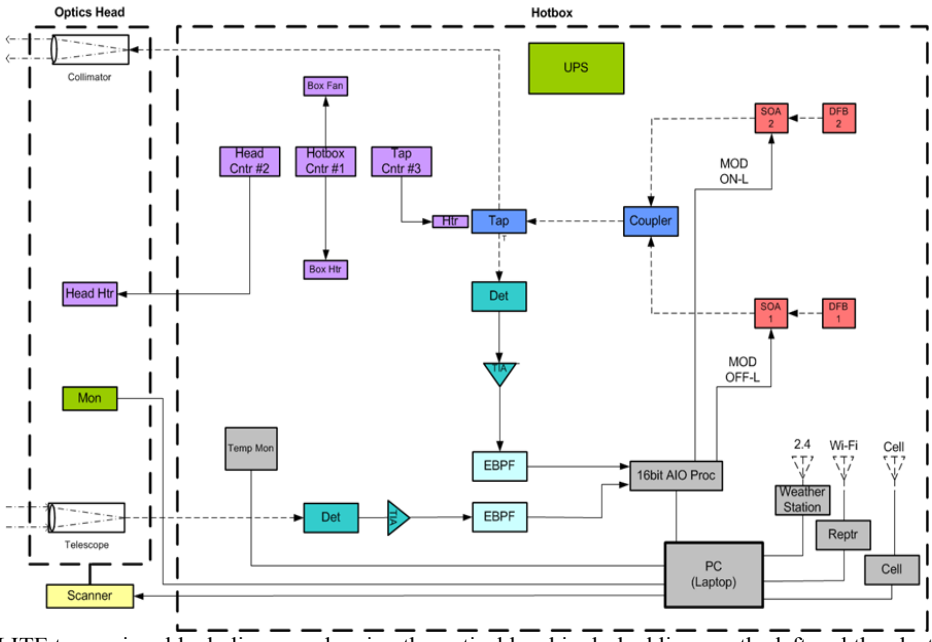


Figure 1. GreenLITE transceiver block diagram, showing the optical head in dashed lines on the left and the electro-optical box on the right.

The GreenLITE transceiver is divided into two main components: 1) the electro-optical box and 2) the optical head. The electro-optical box is a thermally stabilized box which houses the laser transmitters, optical modulators, detector chain

for reference and science channels, computer, data acquisition, and other associated electronics. The optical head is also thermally controlled and houses both transmit and receive telescopes. The telescope box is mounted on a mechanical scanner that is used to point the telescope head at individual retroreflectors. The system is designed to operate in remote locations for long periods of time with minimal interaction.

The current system operates on a line near 1571.112 nm for the online and 1571.062 nm for the offline. The 1 km system transmits a total of 5 mW of optical power combined between online and offline, and the 5 km system uses ~25 mW of transmitted power. Both systems use standard InGaAs detectors and have a transmit divergence of 1.6 mrad, a receiver field of view of 2.6 mrads, and a 25.4 mm transmit telescope. The receiver uses a 25.4 mm telescope for the 1 km system, while a 152.4 mm telescope is used for the 5 km system.

Two transceivers and several retroreflectors are arranged such that the lines (or chords) between the two transceivers and each retroreflector generate a grid of interweaved segments. Each transceiver scans to the retroreflectors in sequence and makes a measurement of each chord for a specific amount of time (nominally 10 seconds). After measuring all of the chords, the transceiver makes a zero path measurement used to evaluate and correct for drift. For the current implementation the maximum scan angle is ~170 degrees, limited mainly by cabling. A conceptual example is given in Figure 2. Note in this example the system is shown in a semi-ellipse configuration which maximizes the use of the retroreflectors by both transceivers, while covering more area than a simple rectilinear configuration (e.g. 31 km^2 versus 25 km^2 for a 5 km chord length). Other optimizations are possible in order to provide the maximum number of intersections over primary areas of interest, or additional transceivers can be used to obtain more uniform coverage and other desired shapes depending on the application. The system requires a clear line of site from the transceivers to the retroreflectors, which can pose challenges in some applications. However, the system testing to date has shown that the algorithms discussed in section 3 have proven robust to a number of non-optimal site layouts and to a limited number of chords being obstructed. This finding enables GreenLITE to be considered for a number of applications where the optimal layout may not be practical, such as an urban environment where the layout will be driven by suitable buildings and structures for mounting the retroreflectors.

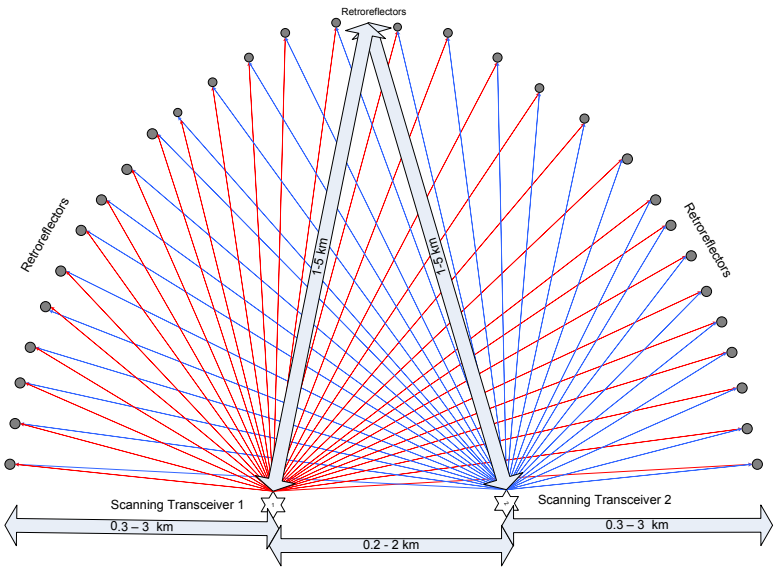


Figure 2. Conceptual diagram of one possible GreenLITE system layout for maximizing spatial coverage.

All hardware control and monitoring, data acquisition, preliminary data processing, and data transmission to the server are performed by the LabVIEW software running on each transceiver. The transceiver software controls the movement of the mechanical scanner and coordinates the acquisition of data with the scanner movement. The software provides a convenient way to load a list of pre-defined retroreflector locations. The actual measurement of the differential transmission of the lasers is made using a digital lockin processing technique. Each laser is intensity modulated at a unique frequency prior to transmission through the atmosphere. A small portion of the laser energy is diverted to a

reference detector whose digitized output can be used to normalize the return signal for variations in transmitted power. The collected laser signal returned by the retroreflectors is digitized and mixed with the local oscillators that were used to modulate the lasers and then normalized by the reference signals. This lockin technique allows very small signals at known frequencies to be detected in the presence of wideband noise many orders of magnitude higher in power than the signal of interest. The normalized online signal is divided by the normalized offline signal to get the differential transmission of the lasers. After each collection period from a retrorflector is complete, the measured differential transmission and all corresponding metadata are formatted and transmitted over the internet to the remote GreenLITE database.

2.2 GreenLITE Data Processing

The GreenLITE system computes near-real-time two-dimensional maps of CO_2 concentrations and estimated fluxes based on a multi-stage approach. This process continuously: 1) computes integrated column CO_2 concentration (ppm) from observed differential laser absorption measurements along horizontal lines of sight between a transceiver and reflectors ("chords"), 2) estimates 2-D distribution of time-varying CO_2 concentrations based on an analytic plume model and an ensemble set of temporally adjacent integrated chord values, and 3) provides modeled flux maps for the geographic extent defined by the chord boundaries via simple temporal differentiation. Currently, this process has been implemented in modular fashion where the applications defined above run asynchronously in a multi-threaded environment coupled to one another through a common database structure. Near real-time monitoring of CO_2 concentrations and 2-D distribution estimates are made possible through a web-based analysis tool.

2.2.1 Retrieving CO_2 Concentration from Observed Optical Depth Ratios

The first step in the retrieval process is the conversion of each GreenLITE laser-based transceiver differential absorption measurement to an estimate of the horizontal column CO_2 concentration in parts per million (ppm). This is accomplished by combining the difference in observed "on" and "off" line optical depth values ($\Delta\tau = \tau_{on} - \tau_{off}$), measured meteorological state, and static path length parameters with a radiative transfer (RT) based iterative retrieval scheme depicted in Figure 3.

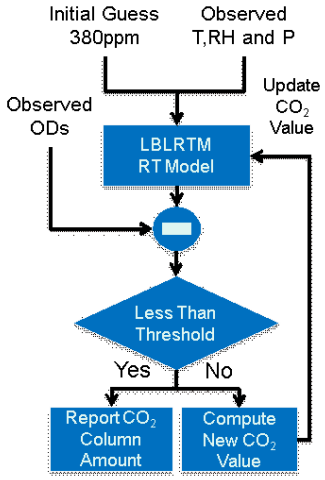


Figure 3. Iterative flow diagram of CO_2 concentration retrieval algorithm

This algorithm employs a line-by-line RT model, LBLRTM [16], in conjunction with a standard steepest descent search technique to find the modeled $\Delta\tau$ values that best match the observed $\Delta\tau$ given the observed atmospheric state (surface temperature, relative humidity (RH), and pressure). First, an initial baseline estimate of the column concentration, nominally 380 ppm, is combined with collocated *in situ* measurements of the surface air temperature, surface water vapor, and air pressure at the height of the transceivers/reflectors along with static parameters describing the chord geometry in the RT modeling framework to compute a baseline estimate of the modeled $\Delta\tau_M$ value defined as

$$\Delta\tau_M = \tau_{Mon} - \tau_{Moff} \quad (1)$$

where τ_{Mon} is the model optical depth at the monochromatic online wavelength, and τ_{Moff} is that at the offline wavelength. LBLRTM optical depths are computed from Voigt line shape functions at atmospheric levels and with a continuum model that includes self- and foreign-broadened water vapor as well as continua for CO_2, O_2, N_2, O_3 and extinction due to Rayleigh scattering. The version used in the work included 2012 updates to the CO_2 line parameters and coupling coefficients based on the work of Devi et al. [17] [18], the O_2 line parameters based on HITRAN [19], and additional quadrupole parameters between 7571 - 8171 cm^{-1}.

Next, the observed $\Delta\tau$ is compared to the model value ($\Delta\tau_M$) and the resulting difference is used to update the estimated column concentration. This is achieved by computing a gradient or the change in column concentration as a function of a change in observed optical depth via finite differences. In this work, the gradient is defined as

$$\nabla CO_2 = \frac{\Delta CO_2}{\Delta\tau_M - \Delta\tau_{M+1}} \quad (2)$$

where $\Delta\tau_{M+1}$ is the modeled differential optical depth computed using LBLRTM and an enhanced column CO_2 value which is the last estimated CO_2 value plus a constant offset ΔCO_2. Currently ΔCO_2 is set to a fixed 2 ppm enhancement over the previously estimated value. Finally, an updated estimate of the column concentration is computed based on

$$CO_{2(n+1)} = CO_{2(n)} + \nabla CO_2(\Delta\tau_M - \Delta\tau) \quad (3)$$

where $CO_{2(n)}$ represents the previously estimated column concentration value, and $CO_{2(n+1)}$ is the updated value. This iterative process is continued until the absolute difference, $|\Delta\tau_M - \Delta\tau|$, falls below the instrument noise threshold or the number of iterations exceeds a user-defined maximum. Nominally, this method converges in one to two iterations under a wide variety of atmospheric and environmental conditions due to the fact that the relationship between observed differential optical depth and CO_2 absorption, assuming a constant atmosphere state over the path lengths of interest, is nearly linear Figure 4 shows a representative set of collocated *in situ* point measurements and the corresponding retrieved chord values. The averaged retrieved values track *in situ* measurements over an extended period of ~18 hours. Some variability is expected due to the fact that the *in situ* data represent a time series for a single point in space, and the retrieved values provide an estimate of the average value over the length of the chord.

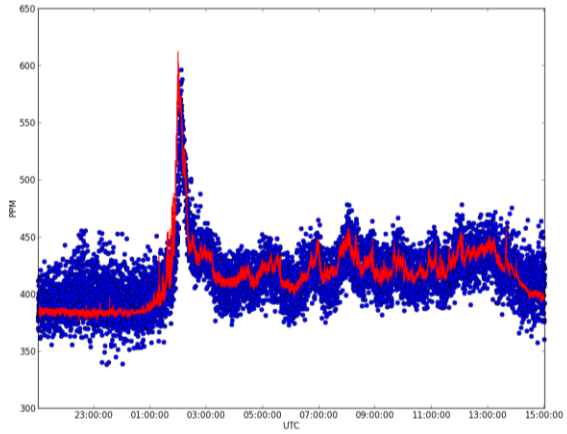


Figure 4 Representative comparison of retrieved column CO_2 values (blue) and collocate in situ field measurements (red). 09/06/2014 15Z

2.2.2 Estimating 2-D Fields of CO_2 Concentrations from Tomographic Samples

The retrieved CO_2 concentration values are then employed in an additional minimization scheme that constructs two-dimensional models of the underlying field concentrations. In traditional tomographic applications, the number of back projections (chords) and angles should optimally approach or exceed the number of pixel elements in the resulting 2-D image, and often the scan geometry is optimized where possible to ensure high-fidelity, unconstrained reconstructions. In the GreenLITE case, the field is under-sampled and the scan pattern is limited due to the number of deployed transceivers and reflectors, site topography, and both natural and man-made barriers. While a number of approaches have been proposed and implemented for a variety of similar applications [20] [21] [22] to optimally constrain this type of problem, the current baseline method models the underlying field concentrations as the sum of a set of analytical functions that are designed to describe the underlying CO_2 background distribution as well as a set of plume-like source terms. While a number of functional forms have been assessed as part of this work, the following analytical model describes the baseline 2-D field contained within the transceiver chord boundaries:

$$\mathrm{F}_{CO_2}(\mathrm{x},\mathrm{y}) = a + bx + cy + dxy + \sum_{n=0}^{N} \alpha_n e^{-\beta_n {x_r}^2} e^{-\gamma_n {y_r}^2} \tag{4}$$

where F(x,y) is the resulting CO_2 concentration as a function of the x/y location, and the linear model on the right-hand side describes the field elements. The first 4 terms in equation 4 describe a basic background term that has a constant offset and is allowed to vary linearly as a function of x and y. The variable ***a*** describes the average background concentration, and the remaining elements describe a simple linear gradient across the field. The summation represents a set of simple Gaussian plumes to describe potential localized CO_2 sources. Each 2-D Gaussian is a function of a normalized set of x_r and y_r parameters defined as

$$x_r = \sqrt{(x - x_n) + (y - y_n)}\cos(\theta_n) \tag{5}$$

and

$$y_r = \sqrt{(x - x_n) + (y - y_n)}\sin(\theta_n) \tag{6}$$

where (x_n,y_n) is the plume center, and θ_n is the angle of rotation. The number of chords limits the number of supportable modeled plumes. The upper limit is defined as the number of chords minus the four background parameters divided by the number of parameters. In this case each plume is defined by six parameters α_n, β_n, γ_n, x_n, y_n and θ_n. For example a site with 42 intersecting chords limits the number of plumes to a maximum of 6 ([42 – 4]/6 = 6.3). In practice, the number of supportable plumes is on the order of 3-4 due to the ability of the minimization technique to uniquely assign values to each element. F(x,y) is constructed using an optimization scheme that minimizes the difference between retrieved CO_2 ppm values and modeled values constructed through discrete integration of F(x,y). This is achieved by minimizing the sum of the root mean square error (RMSE) differences between these values using a sequential least squares programming (SLSQP) optimization algorithm. SLSQP provides a standardized framework for minimizing a user-defined cost function given a set of additional parametric constraints. In this instance the cost function was defined as the sum of the squared differences between a set of observed chord values and modeled chord values created by averaging equally spaced samples along a given chord set for the model described in Equation 4. An additional weighting function was also placed on this overall metric so that the ratio of β_n to γ_n did not exceed some maximum, ensuring that plumes irrespective of direction had some finite width. As part of the overall minimization scheme, the model parameters were constrained to 1) ensure that the plumes were positive in nature and represented source terms and not sinks, 2) to confine plume source locations to the enclosed sampling space, and 3) to ensure plume source amplitudes were within physically realizable bounds for CO_2 emissions.

Figure 5 illustrates the results from a typical reconstruction based on synthetic data. In this example, the image on the left illustrates a typical synthetic field containing 2 centrally located Gaussian plumes on a constant background. The red lines indicate the chords associated with a typical two transceiver configuration with 10 reflectors on each of three sides. Next, the average concentration for each chord is computed based on discrete samples along each chord length. The plot in the center illustrates a simple back projection of the synthetic chord samples. Each dot in the resulting 2-D plot denotes the average value of all chord values that intersect the point of interest. Finally, the panel on the right describes

the resulting reconstruction based on the synthetic data derived from the left hand panel and the reconstruction algorithm described above. In a well-defined environment, this tomographic reconstruction method provides a robust mechanism for estimating well-defined plume-like sources.

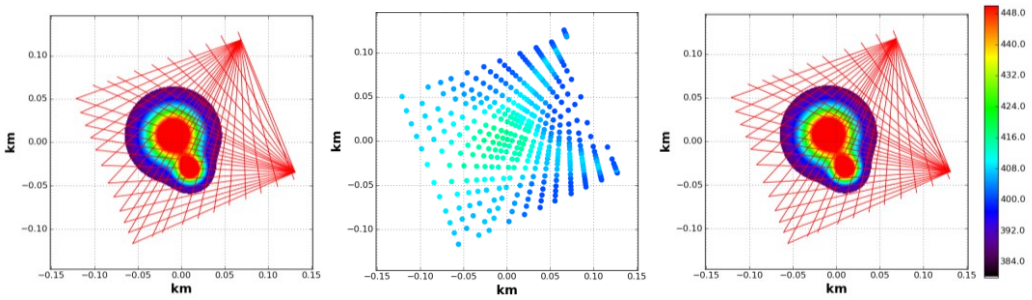


Figure 5 Simulated 2-D reconstruction. Image on the left depicts a synthetic scene and simulated chords for a typical two-transceiver configuration. The center plot depicts a simple back projection of the synthetic chord samples constructed from the field on the left, and the right hand panel illustrates the resulting reconstruction.

2.2.3 Computing 2-D Estimates of CO_2 Flux

With reconstructed 2-D CO_2 fields in hand, a CO_2 flux field can be calculated to characterize the change in CO_2 concentration over time within the physical detection space. The CO_2 flux field (f_{CO2}) is calculated as the delta between chronologically consecutive CO_2 fields or current concentrations and an established background as shown below.

$$f_{CO_2}(x,y) = \frac{F_{CO_2}(x,y) - F'_{CO_2}(x,y)}{\Delta t} \tag{7}$$

where f_{CO2} is the estimated flux, F_{CO2} is current retrieved CO_2 field values as a function of x/y, and at F'_{CO2} is the CO_2 field distribution for some prior time interval or the long-term average/minimum background field and Δt defines the rate of change time interval. While this method is incapable of appropriately differentiating changes from externally derived sources from within the field of interest, it does provides a mechanism for detecting changes in field contributions due to modeled source contributions. An example set of estimated flux values for a set of simulated field trials is shown in Figure 6.

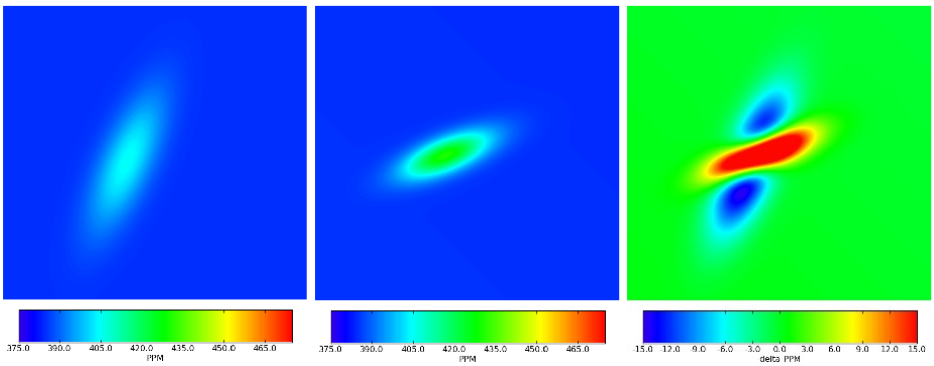


Figure 6. Examples of estimated flux values for simulated fields. Left: Initial simulated plume field at time *t*. Middle: Plume field at time *t*+4 minutes and wind vector rotated clockwise by 45 degrees. Right: Resulting flux image calculated as delta of left and middle images, demonstrating how the plume source has changed over time.

2.2.4 Web-Based Analysis Tool

The GreenLITE system utilizes cloud based 24/7 data collection and processing and a secure web-based user interface. The data from GreenLITE is uploaded remotely to a database hosted on an Amazon Web Server (AWS). The interface

allows direct data access, real-time site monitoring, instrument health monitoring, and provides raw optical depth measurements, retrieved CO_2 concentrations, a near-real-time geolocated map of the spatial distribution of CO_2, and associated weather data including local temperature, pressure, relative humidity, wind speed and direction. An example of this web interface is given in Figure 7. The web tool is readily customizable for new sites and transceiver/reflector configurations.



Figure 7. Examples of the user interface: latest 2D reconstruction and chord data (left), statistics over 24 hour period (center), and weather data (right).

3 TESTING RETRIEVAL OF 2-D CONCENTRATIONS

The system described above was deployed at two sites over an 8-month period in a number of configurations for this evaluation. These deployments include installations at a local farm test facility in Fort Wayne, IN, and the Zero Emissions Research and Technology (ZERT) facility at Montana State University in Bozeman, MT [7].

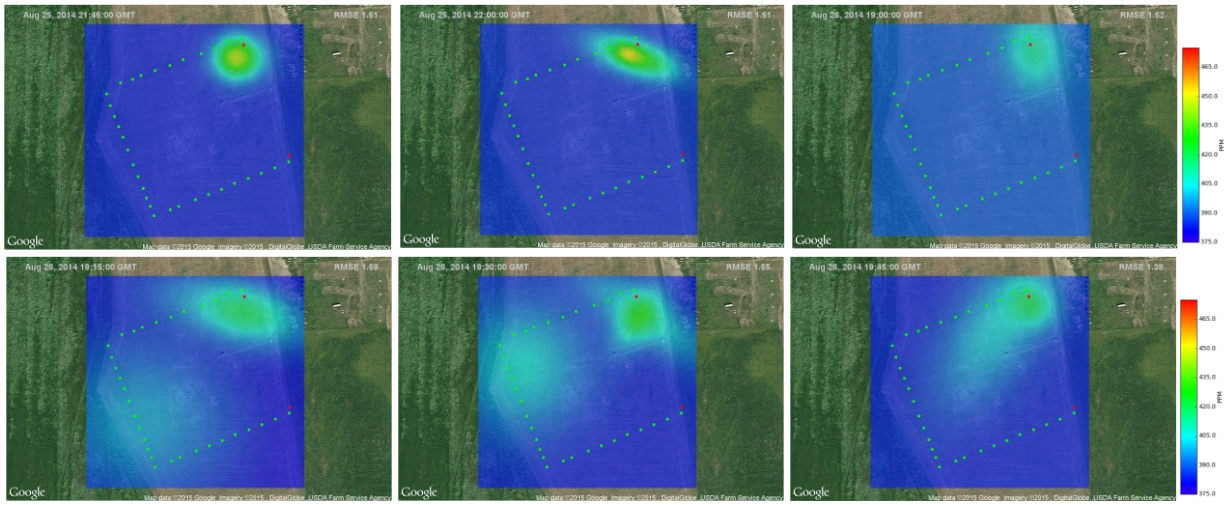


Figure 8. Example site layout for the Zero Emission Research Test Facility (ZERT) and typical time series of 2-D field CO_2 concentrations. Retro reflectors are denoted by green dots and red dots indicate transceiver locations.

Figure 8, Figure 9, and Figure 10 illustrate representative results from a sample set of test cases taken both at the farm test facility in Fort Wayne and the ZERT facility. Figure 8 provides a representative set of reconstructions for the ZERT installation. The green dots in each panel denote the location of the reflectors, and the red dots indicate the geographic location of the transceivers. The predominant feature in the upper right hand corner of each reconstruction correlates well with the presence of a manure pile just off to the right of the upper corner of the reconstructed field. The secondary, more diffuse field corresponds reasonably well to an extended underground controlled release in the lower left hand quadrant of the field inscribed by the reflectors.

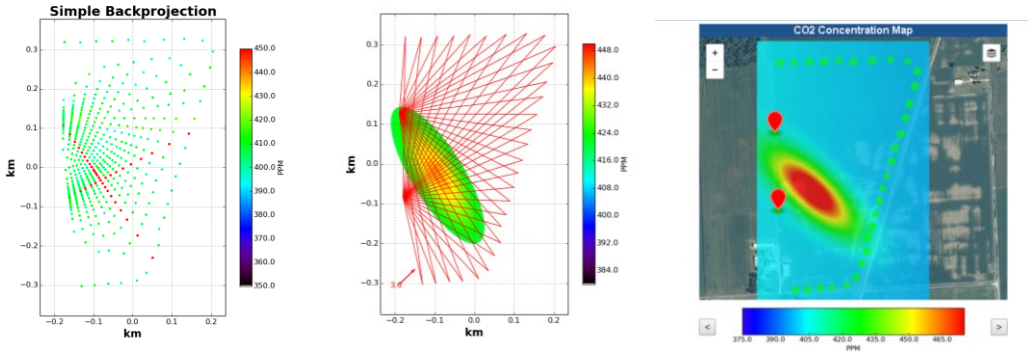


Figure 9. Example of emulated plume data created by pumping CO_2 into short tubes at the farm test site on two of the 60 possible chords. Left: Simple back projection of all chord samples for the time period of interest and the expected chord intersections. Middle: Resulting reconstruction that demonstrates system's capability to detect and locate a plume source. Right: The reconstruction as displayed on the web-based tool during the experiment with background gradient and map.

Figure 9 shows the results of a simulated plume test performed at the farm test facility. One of the challenges in modeling two-dimensional fields of CO_2 concentrations is one's ability to cross-calibrate and/or assess the accuracy of the resulting modeled field values. While numerous methods exist for acquiring *in situ* point measurements of CO_2 concentration as well as flux, it becomes extremely difficult to acquire enough temporally collocated data to accurately represent the distribution over an extended open path field. One can either densely sample the desired open space with numerous similar instruments or move a select set of instruments around the field of interest. Both methods are challenging at best. Having a dense network of similar instruments may be cost prohibitive and would present a calibration problem due primarily to the need to calibrate all the instruments with a common reference before deployment. Moving a single instrument or a small number of instruments around an open field assumes that the distribution is stationary over some extended period in time and that moving the instrument around does not influence the flow dynamics over the area of interest. So in order to test the ability of the GreenLITE system to at least detect the presence of plume-like sources in a cost effective fashion, a set of experiments were devised to enhance concentrations along several chords by placing open ended tubes in front of a select set of reflectors (nominally two) and injecting CO_2 into them to simulate intersecting chords with enhanced concentration values. The results from one such study are illustrated in Figure 9. Two chords were artificially enhanced by the placement of the open-ended tubes along T1R24 with a flow rate of ~1.05 cfm and T2R15 with a flow rate of ~0.75 cfm, where the T denotes the transceiver number top to bottom and the R describes the retroreflector number starting from the top and working clockwise. The retrieved values for these two chords were ~1000 and 750 ppm for the chords T1R24 and T2R15 respectively. The other retrieved values are associated with the ambient field (~390-400 ppm). The plot on the left shows a simple back projection of the chord values over the sampled space. The dots represent the intersections of individual chords and their color values represent the sum of the two retrieved concentration values at each intersection. The intersection of the two primary enhanced chords should notionally indicate the position of the source term. The panel in the center provides the resulting reconstructed field with a single dominant source, while the right panel shows the reconstruction overlaid on the background concentration. The elongation is due to the greater enhancement of the chord along the NW to SE, due to the higher flow rate, which elevates other intersections along that chord. While the model is by no means a perfect representation of an idealized point plume, it does illustrate the ability of the system to detect the presence of a source in automated fashion, and in this case the model located the source to better than 11 meters on an approximately 0.2 km^2 grid while in the presence of ambient CO_2 concentrations and prevailing local wind.

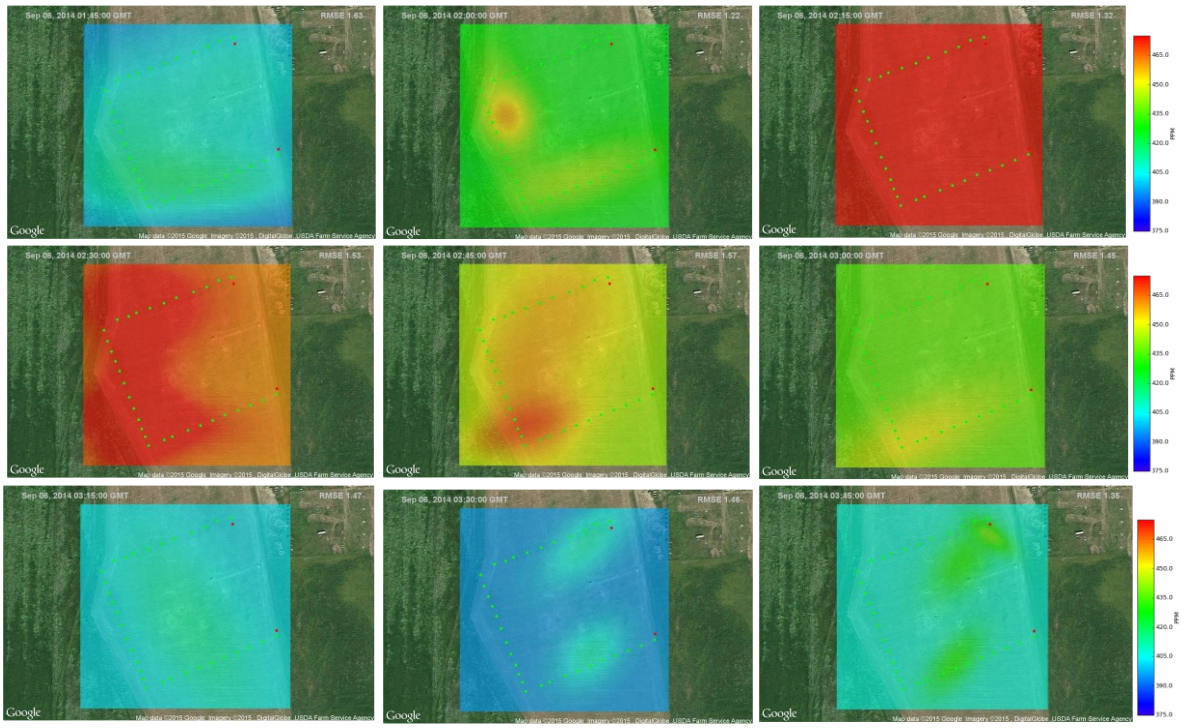


Figure 10 Sample night time reconstructions based on ZERT facility test data collected over same period in time as chord values illustrated in Figure 4. The retro reflector positions are denoted by the green dots in each image, and the red dots indicate the transceiver locations. The images span the time interval between 1:45 and 3:45 GMT on Sept 06, 2014.

Finally, Figure 10 illustrates the model's ability to adapt to changes in background behavior. The ZERT reconstructions shown in this set of panels follow a portion of the time sequence shown in Figure 4. The provided retrievals are centered around the observed spike at dusk (2:15-2:30 GMT) and illustrate how the model tracks changes in background level while still preserving persistent modeled features.

4 CONCLUSIONS

A new measurement capability has been developed that can provide near-real-time information on the spatial distribution of greenhouse gases over geographical areas significant for a number of applications. Field tests have shown that the GreenLITE system measured average CO_2 concentrations that agreed with an independent LI-COR based system in the field of regard of GreenLITE to within a 0.933 correlation coefficient over several days, and as high as 0.98 over one 23 hour continuous period. At the ZERT site we were able to clearly identify signatures from both the buried pipeline during release rates as low as 0.2 tons/day as well as a large natural source from decomposition of a manure pile. Measurement of a controlled release of CO_2 along known chords during a field test resulted in a source detection located to better than 11 meters on a 0.2 km^2 grid while in the presence of ambient CO_2 concentrations and prevailing local wind. While additional work is needed to determine the system's minimum detectability and the timelines required to attribute a persistent feature to a non-natural source, the field tests and operational deployments described here demonstrate GreenLITE's ability to accurately measure CO_2 concentrations and detect/locate CO_2 sources over an extended geographic area.

We focused mainly on one application for the GreenLITE system, monitoring of GCS facilities, which was demonstrated by multiple field campaigns with controlled releases and operationally over an extended 6 month deployment to a ground carbon sequestration site (to be published). In addition, Harris and AER have expanded the GreenLITE concept based on our GCS work and have now developed and begun testing a system that will allow us to examine additional applications such as direct urban plume measurements. The updated system will undergo tests in Boulder, CO, in September 2015, and will then be deployed to an urban environment for operational evaluation. Other potential

applications may include oil and gas productions sites, large facility monitoring, and scientific evaluation of natural and anthropogenic sources and sinks.

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